# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Audit Indicator | Global Session Mean (± Std) |
|---|---|
| Synthetic Fidelity (JSD) | 91.5 ± 1.63 |
| Correlation Stability | 93.9 ± 0.83 |
| Distribution Consistency | 92.4 ± 1.43 |
| Entropy Preservation | 94.9 ± 0.54 |
| Mean Z Score Drift | 0.1 ± 0.02 |
| Categorical Fidelity | 0.0 ± 0.00 |
| Privacy Risk | 0.2 ± 0.40 |
| Data Density | 100.0 ± 0.00 |
| Coverage Score | 98.4 ± 0.92 |
| Feature Accuracy | 93.8 ± 1.33 |
| Novelty Score | 0.8 ± 0.01 |

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## Dataset Audit: heart_failure_clinical_records.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 299 |
| Columns | 13 |
| Numeric features | 13 |
| Categorical features | 0 |
| Missing cells (%) | 0.0% |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | gpt-4o-mini |
| Folds (*13) | 10 |
| Context % (*14) | 0% |
| Generation % (*15) | 100% |
| Join context (*16) | False |

### 3. Statistical Significance & Data Drift

| Feature | Orig Mean | Aug Mean | Shift | P-Value | Status |
|---|---|---|---|---|---|
| age | 60.834 | 60.331 | -0.503 | 0.8% | OK |
| anaemia | 0.431 | 0.456 | +0.025 | 5.7% | OK |
| creatinine_phosphokinase | 581.839 | 555.728 | -26.111 | 4.5% | OK |
| diabetes | 0.418 | 0.477 | +0.059 | 14.1% | Warn |
| ejection_fraction | 38.084 | 38.293 | +0.209 | 0.5% | OK |
| high_blood_pressure | 0.351 | 0.364 | +0.013 | 3.7% | OK |
| platelets | 263358.029 | 263302.656 | -55.374 | 0.0% | OK |
| serum_creatinine | 1.394 | 1.413 | +0.019 | 1.4% | OK |
| serum_sodium | 136.625 | 136.485 | -0.140 | 0.1% | OK |
| sex | 0.649 | 0.640 | -0.009 | 1.3% | OK |
| smoking | 0.321 | 0.339 | +0.018 | 5.6% | OK |
| time | 130.261 | 139.213 | +8.953 | 6.9% | OK |

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Synthetic Fidelity (JSD) (*1) | 91.5 ± 1.63 |
| Correlation Stability (*2) | 93.9 ± 0.83 |
| Distribution Consistency (*3) | 92.4 ± 1.43 |
| Entropy Preservation (*4) | 94.9 ± 0.54 |
| Mean Z Score Drift (*5) | 0.1 ± 0.02 |
| Categorical Fidelity (*6) | 0.0 ± 0.00 |
| Privacy Risk (*7) | 0.2 ± 0.40 |

| | |
|---|---|
| Data Density (*8) | 100.0 ± 0.00 |
| Coverage Score (*9) | 98.4 ± 0.92 |
| Feature Accuracy (*10) | 93.8 ± 1.33 |
| Novelty Score (*11) | 0.8 ± 0.01 |

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

*[6] Synthetic Fidelity (JSD): Overall distribution similarity.*
*[7] Correlation Stability: Multivariate relationships preserved.*
*[8] Distribution Consistency: Median feature similarity.*
*[9] Entropy Preservation: Information diversity maintained.*
*[10] Mean Z Score Drift: Standardized distribution shift.*
*[11] Categorical Fidelity: Category distribution similarity.*
*[12] Privacy Risk: Synthetic rows too close to originals.*
*[13] Data Density: Non-missing cells in augmented data.*
*[14] Coverage Score: Synthetic coverage of original space.*
*[15] Feature Accuracy: Mean feature similarity.*
*[16] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

### Model: GB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8300000000000001 | 0.8066666666666666 | 0.8200000000000001 | -0.010000000000000009 | |
| F1 | 0.7532401666570763 | 0.7139515793018381 | 0.7379387779387779 | -0.015301388718298236 | |

### Model: XGB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.835 | 0.8150000000000001 | 0.8233333333333333 | -0.011666666666666681 | |
| F1 | 0.7547418227770806 | 0.7249623403425496 | 0.7407567796065199 | -0.013985043170560707 | |

### Model: DT

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7966666666666666 | 0.7533333333333332 | 0.7533333333333333 | -0.04333333333333332 | |
| F1 | 0.6966579947551444 | 0.6339883342373585 | 0.6299602830413119 | -0.06669771171383268 | |

### Model: NB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7499999999999999 | 0.7333333333333333 | 0.7433333333333334 | -0.006666666666666667 | |
| F1 | 0.5667171896628354 | 0.545460496416765 | 0.5593253556352261 | -0.00739183402760922 | |

### Model: SVM

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6333333333333334 | 0.6333333333333334 | 0.6333333333333334 | 0 | |
| F1 | N/A | 0.0 | N/A | 0 | |

### Model: AB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| ACCURACY | 0.825 | 0.8150000000000001 | 0.8183333333333336 | -0.006666666666666665 | |
| F1 | 0.7417404771442427 | 0.7240577447448926 | 0.7326994823202254 | -0.009040994824017379 | |

### Model: RF

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8300000000000001 | 0.8233333333333335 | 0.8299999999999998 | -1.1102230246251566e-17 | |
| F1 | 0.7390976630511515 | 0.7293475433814793 | 0.7432717283487 | 0.004174065297548624 | |

### Model: LR

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8066666666666666 | 0.8016666666666665 | 0.8066666666666666 | 2.2204460492503132e-17 | |
| F1 | 0.6937863401621888 | 0.6903764940622562 | 0.6993114547742298 | 0.005525114612041015 | |

### Model: MLP

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.565 | 0.4649999999999997 | 0.5566666666666666 | -0.008333333333333326 | |
| F1 | 0.17458245519997076 | 0.2849062511562512 | 0.22909364328618026 | 0.05451118808620951 | |

### Model: KNN

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6133333333333335 | 0.6300000000000001 | 0.6250000000000001 | 0.011666666666666659 | |
| F1 | 0.007142857142857143 | 0.1835940590498795 | 0.13434720183547383 | 0.12720434469261668 | |

## Appendix: Benchmark Data Trace

*Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline*

### A. Baseline: Original Dataset (Full Record Sample)

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 75.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 265000.0 | 1.9 | 130.0 | 1.0 |
| 2 | 55.0 | 0.0 | 7861.0 | 0.0 | 38.0 | 0.0 | 263358.03 | 1.1 | 136.0 | 1.0 |
| 3 | 65.0 | 0.0 | 146.0 | 0.0 | 20.0 | 0.0 | 162000.0 | 1.3 | 129.0 | 1.0 |
| 4 | 50.0 | 1.0 | 111.0 | 0.0 | 20.0 | 0.0 | 210000.0 | 1.9 | 137.0 | 1.0 |
| 5 | 65.0 | 1.0 | 160.0 | 1.0 | 20.0 | 0.0 | 327000.0 | 2.7 | 116.0 | 0.0 |
| 6 | 90.0 | 1.0 | 47.0 | 0.0 | 40.0 | 1.0 | 204000.0 | 2.1 | 132.0 | 1.0 |
| 7 | 75.0 | 1.0 | 246.0 | 0.0 | 15.0 | 0.0 | 127000.0 | 1.2 | 137.0 | 1.0 |
| 8 | 60.0 | 1.0 | 315.0 | 1.0 | 60.0 | 0.0 | 454000.0 | 1.1 | 131.0 | 1.0 |
| 9 | 65.0 | 0.0 | 157.0 | 0.0 | 65.0 | 0.0 | 263358.03 | 1.5 | 138.0 | 0.0 |
| 10 | 80.0 | 1.0 | 123.0 | 0.0 | 35.0 | 1.0 | 388000.0 | 9.4 | 133.0 | 1.0 |
| 11 | 75.0 | 1.0 | 81.0 | 0.0 | 38.0 | 1.0 | 368000.0 | 4.0 | 131.0 | 1.0 |
| 12 | 62.0 | 0.0 | 231.0 | 0.0 | 25.0 | 1.0 | 253000.0 | 0.9 | 140.0 | 1.0 |
| 13 | 45.0 | 1.0 | 981.0 | 0.0 | 30.0 | 0.0 | 136000.0 | 1.1 | 137.0 | 1.0 |
| 14 | 50.0 | 1.0 | 168.0 | 0.0 | 38.0 | 1.0 | 276000.0 | 1.1 | 137.0 | 1.0 |
| 15 | 49.0 | 1.0 | 80.0 | 0.0 | 30.0 | 1.0 | 427000.0 | 1.0 | 138.0 | 0.0 |
| 16 | 82.0 | 1.0 | 379.0 | 0.0 | 50.0 | 0.0 | 47000.0 | 1.3 | 136.0 | 1.0 |
| 17 | 87.0 | 1.0 | 149.0 | 0.0 | 38.0 | 0.0 | 262000.0 | 0.9 | 140.0 | 1.0 |
| 18 | 45.0 | 0.0 | 582.0 | 0.0 | 14.0 | 0.0 | 166000.0 | 0.8 | 127.0 | 1.0 |
| 19 | 70.0 | 1.0 | 125.0 | 0.0 | 25.0 | 1.0 | 237000.0 | 1.0 | 140.0 | 0.0 |
| 20 | 48.0 | 1.0 | 582.0 | 1.0 | 55.0 | 0.0 | 87000.0 | 1.9 | 121.0 | 0.0 |

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## B.1 Audit Trace: Fold 1 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 70.0 | 0.0 | 81.0 | 1.0 | 35.0 | 1.0 | 229000.0 | 1.3 | 132.0 | 1.0 |
| 2 | 67.0 | 1.0 | 646.0 | 1.0 | 25.0 | 1.0 | 231000.0 | 1.1 | 138.0 | 0.0 |
| 3 | 90.0 | 1.0 | 337.0 | 0.0 | 40.0 | 0.0 | 321000.0 | 1.9 | 131.0 | 1.0 |
| 4 | 60.0 | 1.0 | 754.0 | 1.0 | 35.0 | 1.0 | 179000.0 | 1.18 | 145.0 | 1.0 |
| 5 | 85.0 | 0.0 | 224.0 | 0.0 | 35.0 | 0.0 | 262000.0 | 0.9 | 140.0 | 1.0 |
| 6 | 63.0 | 1.0 | 103.0 | 0.0 | 60.0 | 1.0 | 267000.0 | 1.0 | 141.0 | 0.0 |
| 7 | 45.0 | 1.0 | 66.0 | 1.0 | 25.0 | 0.0 | 233000.0 | 0.8 | 135.0 | 1.0 |
| 8 | 70.0 | 0.0 | 161.0 | 0.0 | 25.0 | 0.0 | 244000.0 | 1.2 | 142.0 | 0.0 |
| 9 | 58.0 | 1.0 | 400.0 | 0.0 | 40.0 | 0.0 | 164000.0 | 1.0 | 139.0 | 0.0 |
| 10 | 55.0 | 0.0 | 572.0 | 1.0 | 35.0 | 0.0 | 231000.0 | 0.8 | 143.0 | 0.0 |

## B.2 Audit Trace: Fold 2 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 45.0 | 0.0 | 2442.0 | 0.0 | 17.0 | 1.0 | 422000.0 | 1.1 | 138.0 | 0.0 |
| 2 | 42.0 | 0.0 | 7702.0 | 0.0 | 14.0 | 0.0 | 222000.0 | 1.83 | 135.0 | 1.0 |
| 3 | 82.0 | 0.0 | 5882.0 | 0.0 | 45.0 | 1.0 | 350000.0 | 2.0 | 136.0 | 1.0 |
| 4 | 82.0 | 0.0 | 23.0 | 1.0 | 30.0 | 1.0 | 200000.0 | 3.5 | 145.0 | 1.0 |
| 5 | 90.0 | 1.0 | 149.0 | 0.0 | 38.0 | 0.0 | 196000.0 | 1.0 | 142.0 | 0.0 |
| 6 | 78.0 | 0.0 | 4540.0 | 1.0 | 35.0 | 0.0 | 263358.03 | 1.2 | 137.0 | 1.0 |
| 7 | 60.0 | 0.0 | 2656.0 | 1.0 | 30.0 | 0.0 | 305000.0 | 2.3 | 137.0 | 1.0 |
| 8 | 45.0 | 1.0 | 130.0 | 0.0 | 35.0 | 0.0 | 174000.0 | 0.8 | 139.0 | 1.0 |
| 9 | 45.0 | 0.0 | 582.0 | 1.0 | 55.0 | 0.0 | 543000.0 | 1.0 | 132.0 | 0.0 |
| 10 | 65.0 | 1.0 | 59.0 | 1.0 | 60.0 | 0.0 | 172000.0 | 0.9 | 137.0 | 0.0 |

## B.3 Audit Trace: Fold 3 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 80.0 | 1.0 | 123.0 | 1.0 | 50.0 | 1.0 | 406000.0 | 1.4 | 133.0 | 1.0 |
| 2 | 42.0 | 0.0 | 64.0 | 0.0 | 40.0 | 1.0 | 174000.0 | 1.1 | 136.0 | 1.0 |
| 3 | 55.0 | 0.0 | 63.0 | 1.0 | 25.0 | 0.0 | 212000.0 | 5.0 | 136.0 | 0.0 |
| 4 | 87.0 | 1.0 | 149.0 | 0.0 | 50.0 | 0.0 | 406000.0 | 2.0 | 134.0 | 1.0 |
| 5 | 80.0 | 0.0 | 582.0 | 0.0 | 25.0 | 1.0 | 127000.0 | 4.4 | 133.0 | 1.0 |
| 6 | 54.0 | 1.0 | 1808.0 | 0.0 | 60.0 | 1.0 | 220000.0 | 0.9 | 141.0 | 0.0 |
| 7 | 50.0 | 0.0 | 582.0 | 1.0 | 38.0 | 0.0 | 310000.0 | 1.9 | 135.0 | 1.0 |
| 8 | 79.0 | 1.0 | 55.0 | 0.0 | 50.0 | 1.0 | 172000.0 | 1.8 | 133.0 | 1.0 |
| 9 | 65.0 | 1.0 | 160.0 | 1.0 | 20.0 | 0.0 | 327000.0 | 2.7 | 116.0 | 0.0 |
| 10 | 70.0 | 0.0 | 92.0 | 0.0 | 60.0 | 1.0 | 317000.0 | 0.8 | 140.0 | 0.0 |

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## B.4 Audit Trace: Fold 4 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 57.0 | 1.0 | 250.0 | 1.0 | 25.0 | 0.0 | 263000.0 | 1.1 | 141.0 | 1.0 |
| 2 | 45.0 | 0.0 | 308.0 | 1.0 | 35.0 | 0.0 | 850000.0 | 1.0 | 139.0 | 0.0 |
| 3 | 60.0 | 0.0 | 2261.0 | 1.0 | 30.0 | 1.0 | 328000.0 | 0.9 | 138.0 | 1.0 |
| 4 | 75.0 | 0.0 | 224.0 | 0.0 | 38.0 | 1.0 | 265000.0 | 0.9 | 137.0 | 1.0 |
| 5 | 60.0 | 1.0 | 754.0 | 1.0 | 45.0 | 0.0 | 297000.0 | 0.8 | 134.0 | 1.0 |
| 6 | 40.0 | 1.0 | 129.0 | 0.0 | 25.0 | 0.0 | 263358.03 | 0.9 | 137.0 | 0.0 |
| 7 | 48.0 | 1.0 | 131.0 | 1.0 | 30.0 | 1.0 | 244000.0 | 1.6 | 130.0 | 0.0 |
| 8 | 95.0 | 1.0 | 371.0 | 0.0 | 30.0 | 0.0 | 461000.0 | 2.0 | 132.0 | 1.0 |
| 9 | 45.0 | 1.0 | 981.0 | 0.0 | 30.0 | 0.0 | 136000.0 | 1.1 | 137.0 | 1.0 |
| 10 | 50.0 | 0.0 | 582.0 | 1.0 | 38.0 | 0.0 | 310000.0 | 1.9 | 135.0 | 1.0 |

## B.5 Audit Trace: Fold 5 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 75.0 | 0.0 | 119.0 | 0.0 | 60.0 | 0.0 | 248000.0 | 0.7 | 139.0 | 1.0 |
| 2 | 65.0 | 1.0 | 62.0 | 0.0 | 20.0 | 0.0 | 70000.0 | 1.0 | 138.0 | 1.0 |
| 3 | 47.0 | 0.0 | 115.0 | 0.0 | 60.0 | 1.0 | 184000.0 | 1.1 | 141.0 | 1.0 |
| 4 | 50.0 | 0.0 | 250.0 | 1.0 | 25.0 | 0.0 | 329000.0 | 1.3 | 136.0 | 1.0 |
| 5 | 73.0 | 0.0 | 675.0 | 0.0 | 40.0 | 0.0 | 279000.0 | 1.4 | 137.0 | 0.0 |
| 6 | 85.0 | 0.0 | 129.0 | 0.0 | 50.0 | 0.0 | 184000.0 | 2.5 | 137.0 | 0.0 |
| 7 | 78.0 | 0.0 | 224.0 | 0.0 | 50.0 | 0.0 | 481000.0 | 1.4 | 138.0 | 1.0 |
| 8 | 70.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 263358.03 | 1.83 | 134.0 | 1.0 |
| 9 | 61.0 | 0.0 | 248.0 | 0.0 | 30.0 | 1.0 | 267000.0 | 0.7 | 136.0 | 1.0 |
| 10 | 40.0 | 0.0 | 478.0 | 1.0 | 30.0 | 0.0 | 303000.0 | 0.9 | 136.0 | 1.0 |

## B.6 Audit Trace: Fold 6 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 70.0 | 1.0 | 75.0 | 0.0 | 38.0 | 1.0 | 153000.0 | 1.0 | 136.0 | 0.0 |
| 2 | 72.0 | 1.0 | 231.0 | 0.0 | 25.0 | 1.0 | 290000.0 | 0.9 | 140.0 | 1.0 |
| 3 | 61.0 | 0.0 | 231.0 | 1.0 | 30.0 | 1.0 | 334000.0 | 0.9 | 142.0 | 1.0 |
| 4 | 65.0 | 0.0 | 936.0 | 1.0 | 45.0 | 0.0 | 395000.0 | 2.1 | 139.0 | 1.0 |
| 5 | 70.0 | 1.0 | 171.0 | 0.0 | 60.0 | 1.0 | 218000.0 | 0.9 | 134.0 | 1.0 |
| 6 | 45.0 | 1.0 | 427.0 | 1.0 | 17.0 | 1.0 | 136000.0 | 2.3 | 128.0 | 1.0 |
| 7 | 81.0 | 0.0 | 4540.0 | 0.0 | 35.0 | 0.0 | 231000.0 | 1.18 | 137.0 | 1.0 |
| 8 | 70.0 | 0.0 | 2695.0 | 1.0 | 40.0 | 0.0 | 241000.0 | 1.0 | 137.0 | 1.0 |
| 9 | 70.0 | 0.0 | 835.0 | 0.0 | 35.0 | 1.0 | 305000.0 | 0.8 | 133.0 | 0.0 |
| 10 | 80.0 | 0.0 | 776.0 | 1.0 | 38.0 | 1.0 | 192000.0 | 1.3 | 135.0 | 0.0 |

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## B.7 Audit Trace: Fold 7 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 70.0 | 0.0 | 161.0 | 0.0 | 35.0 | 1.0 | 289000.0 | 0.9 | 137.0 | 0.0 |
| 2 | 62.0 | 1.0 | 104.0 | 1.0 | 50.0 | 0.0 | 507000.0 | 1.2 | 137.0 | 1.0 |
| 3 | 90.0 | 1.0 | 371.0 | 0.0 | 35.0 | 0.0 | 235000.0 | 0.8 | 141.0 | 1.0 |
| 4 | 45.0 | 1.0 | 981.0 | 0.0 | 60.0 | 0.0 | 263358.03 | 1.2 | 136.0 | 0.0 |
| 5 | 52.0 | 0.0 | 190.0 | 1.0 | 40.0 | 0.0 | 253000.0 | 1.18 | 145.0 | 1.0 |
| 6 | 80.0 | 0.0 | 898.0 | 0.0 | 38.0 | 1.0 | 47000.0 | 1.1 | 137.0 | 0.0 |
| 7 | 75.0 | 0.0 | 582.0 | 1.0 | 30.0 | 1.0 | 263358.03 | 1.83 | 134.0 | 0.0 |
| 8 | 53.0 | 0.0 | 207.0 | 1.0 | 40.0 | 0.0 | 223000.0 | 1.2 | 130.0 | 0.0 |
| 9 | 60.0 | 1.0 | 2281.0 | 1.0 | 40.0 | 0.0 | 283000.0 | 1.0 | 141.0 | 0.0 |
| 10 | 59.0 | 1.0 | 280.0 | 1.0 | 25.0 | 1.0 | 302000.0 | 1.0 | 141.0 | 0.0 |

## B.8 Audit Trace: Fold 8 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 60.0 | 0.0 | 320.0 | 1.0 | 40.0 | 0.0 | 279000.0 | 1.7 | 136.0 | 1.0 |
| 2 | 70.0 | 0.0 | 233.0 | 0.0 | 25.0 | 1.0 | 338000.0 | 0.9 | 141.0 | 0.0 |
| 3 | 50.0 | 0.0 | 124.0 | 1.0 | 25.0 | 1.0 | 284000.0 | 1.1 | 128.0 | 1.0 |
| 4 | 60.0 | 1.0 | 588.0 | 0.0 | 20.0 | 0.0 | 219000.0 | 1.0 | 140.0 | 1.0 |
| 5 | 65.0 | 1.0 | 113.0 | 0.0 | 35.0 | 0.0 | 179000.0 | 1.2 | 139.0 | 1.0 |
| 6 | 58.0 | 1.0 | 129.0 | 1.0 | 40.0 | 0.0 | 150000.0 | 1.0 | 139.0 | 0.0 |
| 7 | 65.0 | 0.0 | 224.0 | 1.0 | 50.0 | 0.0 | 149000.0 | 1.3 | 137.0 | 1.0 |
| 8 | 80.0 | 0.0 | 898.0 | 0.0 | 25.0 | 0.0 | 149000.0 | 1.1 | 144.0 | 1.0 |
| 9 | 66.0 | 1.0 | 72.0 | 0.0 | 40.0 | 1.0 | 242000.0 | 1.2 | 134.0 | 1.0 |
| 10 | 75.0 | 1.0 | 582.0 | 0.0 | 30.0 | 0.0 | 225000.0 | 1.83 | 134.0 | 1.0 |

## B.9 Audit Trace: Fold 9 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 75.0 | 1.0 | 81.0 | 0.0 | 65.0 | 1.0 | 427000.0 | 5.8 | 136.0 | 1.0 |
| 2 | 53.0 | 1.0 | 180.0 | 1.0 | 60.0 | 0.0 | 264000.0 | 1.3 | 145.0 | 0.0 |
| 3 | 63.0 | 1.0 | 84.0 | 0.0 | 38.0 | 1.0 | 507000.0 | 0.9 | 140.0 | 0.0 |
| 4 | 65.0 | 0.0 | 118.0 | 0.0 | 25.0 | 0.0 | 267000.0 | 0.8 | 140.0 | 1.0 |
| 5 | 60.0 | 1.0 | 104.0 | 1.0 | 38.0 | 0.0 | 253000.0 | 0.9 | 136.0 | 1.0 |
| 6 | 58.0 | 0.0 | 582.0 | 0.0 | 62.0 | 0.0 | 263358.03 | 0.8 | 139.0 | 0.0 |
| 7 | 55.0 | 0.0 | 572.0 | 1.0 | 35.0 | 0.0 | 231000.0 | 0.8 | 143.0 | 0.0 |
| 8 | 60.0 | 0.0 | 68.0 | 0.0 | 20.0 | 0.0 | 119000.0 | 2.9 | 127.0 | 1.0 |
| 9 | 70.0 | 0.0 | 97.0 | 0.0 | 60.0 | 1.0 | 220000.0 | 0.9 | 138.0 | 1.0 |
| 10 | 53.0 | 1.0 | 1808.0 | 0.0 | 60.0 | 1.0 | 249000.0 | 0.7 | 138.0 | 1.0 |

# TabuLLM: AI Benchmark Report

## B.10 Audit Trace: Fold 10 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 62.0 | 0.0 | 281.0 | 1.0 | 30.0 | 0.0 | 279000.0 | 1.5 | 145.0 | 1.0 |
| 2 | 75.0 | 1.0 | 582.0 | 0.0 | 38.0 | 1.0 | 263358.03 | 2.5 | 135.0 | 1.0 |
| 3 | 55.0 | 0.0 | 60.0 | 0.0 | 40.0 | 0.0 | 368000.0 | 0.7 | 136.0 | 1.0 |
| 4 | 54.0 | 1.0 | 707.0 | 0.0 | 38.0 | 0.0 | 330000.0 | 1.8 | 135.0 | 0.0 |
| 5 | 44.0 | 0.0 | 582.0 | 0.0 | 30.0 | 1.0 | 213000.0 | 0.8 | 137.0 | 1.0 |
| 6 | 78.0 | 1.0 | 109.0 | 0.0 | 35.0 | 0.0 | 277000.0 | 1.7 | 137.0 | 1.0 |
| 7 | 60.0 | 0.0 | 897.0 | 1.0 | 45.0 | 0.0 | 297000.0 | 1.0 | 133.0 | 1.0 |
| 8 | 75.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 265000.0 | 1.9 | 130.0 | 1.0 |
| 9 | 45.0 | 1.0 | 1876.0 | 1.0 | 35.0 | 0.0 | 226000.0 | 0.9 | 138.0 | 1.0 |
| 10 | 75.0 | 1.0 | 246.0 | 0.0 | 15.0 | 0.0 | 127000.0 | 1.2 | 137.0 | 1.0 |